(Please write your Enrollment Number) Enrollment No. 032

End-Term Examination

(CBCS Regular & Reappear) (SUBJECTIVE TYPE)

Course Name: < B.Tech, (IT,CSE,MAE,ECE,DMAM,ECE-AI) > Semester:< 3 >

(Nov. - Dec, 2022)

Subject Code: BIT - 201 **Subject: Database Management Systems**

Time : 3 Hours **Maximum Marks :** 60

Note: Q. 1 is compulsory. Attempt ONE question each from the Units I, II, III & IV.

Q1. **(2.5*8=20)**

(a) Differentiate between Primary Key, Super Key and Candidate key.
(b) Explain the 3 level schema architecture?
(c) What is a lossless join decomposition?
(d) What are views? How are they created in database? Give example.
(e) Convert the following SQL query into relational algebra expression:

```
Select Count (A1)
From R1
Where A2 > = 10
```

(f) Explain the problem of dirty read with example during Concurrent Transactions.
(g) Explain the ACID properties of a transaction.
(h) When are two relations said to be union compatible? Give example..

UNIT-1

Q2. (a) A music company decides to create a Database having Musician details like Aadhar no, Name, address, phone no. Poorly paid musicians often share the same address, but no address has more than one phone. While recording a song, many musical instruments are used like guitar, synthetic flute etc., these also have a musical key (like C, B-flat, E-flat etc). Each music Album that is recorded has a unique album identifier, title, copyright date, format (CD or MC), no. of songs. Every recorded song has a unique title and an author. Multi -talented musicians can play multiple instruments and one instrument can be shared among several musicians. A musician can produce several albums, but each album has a single producer. Draw an ER diagram for the music company mentioning ALL the constraints and assumptions (if any). **(6+4)**

(b) Differentiate between strong and weak entity set.

Q3. (a) A company database needs to store information about employees (identified by *eid*, with *salary* and *phone* details), departments (identified by *dna*, with *dname* and *budget*), and children of employees (with *name* and *age*). Employees *work* in departments; each department is *managed by* an employee; a child must be identified uniquely by *name* when the parent (who is an employee; assume that only one parent works for the company) is known. We are not interested in information about a child once the parent leaves the company. Draw an ER diagram that captures this information. **(5+5)**

(b) Convert the above ER diagram into relational model mentioning all Key constraints.

UNIT-2

Q4. Consider the following relations containing airline flight information: **(10)**

Flights(fno: integer, from: string, to: string, distance: integer, departs: time, arrives: time) Aircraft(aid: integer, aname: string, Tuisingrange: integer)
Certified(eid: integer, aid: integer)
Employees(eid: integer, ename: string, salary: integer)

Note that the Employees relation describes pilots and other kinds of employees as well; every pilot is certified for some aircraft (otherwise, he or she would not qualify as a pilot), and only pilots are certified to fly.
Answer the following queries using SQL statements.

PTO

a. Give details of employees with salary greater than 20,000.
b. Give the details of flights staring from Delhi.
c. Find aids of all aircraft that can be used on non-stop flights from Delhi to Chennai.
d. Find the names of pilots earning more than 1,00,000 rupees.
e. Find the eids of employees who make the highest salary.
f. Find the total amount paid to employees as salaries

Q5. Consider following relational schema and write the relational algebra expressions. **(10)**
Sailors(sid: integer, sname: string, rating: integer, age: real);
Boats(bid: integer, bname: string, color: string);
Reserves(sid: integer, bid: integer, day: date).

(a) Find the names of sailors who have reserved both a red and a green boat
(b) Find the sids of sailors with age over 20 who have not reserved a red boat.
(c) Retrieve is the color of boat reverse by 'ABC Sailor'?
(d) Find the names of sailors who have never reserved boat number 103

UNIT-3

Q6. Given a relation R(X, Y, Z, W, P) and Functional Dependency set FD = { $X \rightarrow Y$, $Y \rightarrow P$, and $Z \rightarrow W$}, **(10)**

a) Find the closure of F.
b) Find the Primary Key of F.
c) Determine whether the given R is in 3NF? If not convert it into 3 NF.

Q7. (a) Why do we need normalization? Differentiate between 3NF and BCNF. **(5+5)**
(b) What are serial and non-serial schedules? When is a schedule Serializable? Explain with examples.

UNIT-4

Q8. (a) What is a deadlock? How can they be prevented? Give examples. **(5+5)**
(b) Write short notes on Multimedia databases. **(5+5)**

Q9. (a) What do you mean by two phase locking protocols. Explain with example.
(b) Explain the framework of distributed DBMS.